Ashish Kumar | **23B1028**
Computer Science & Engineering | **B.Tech.**
Indian Institute of Technology Bombay | **Gender: Male**
DOB: 04/10/2004

Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2027	7.17 **Core:** 172 **| Total:** 202
Intermediate	CBSE	JNV Kottayam, Kerala	2023	91.00%
Matriculation	CBSE	JNV Jamui, Bihar	2021	82.20%

Scholastic Achievements

- Qualified for Stage-I of the **National Talent Search Examination (NTSE)**, Government of India *(2020)*
- Recognized by **Dakshana Foundation** for academic performance and achievement in competitive exams *(2023)*
- Awarded the **FFE Scholarship** in recognition of academic merit and continued excellence in studies *(2023)*
- Achieved **Specialist** rank on Codeforces through regular participation in competitive programming contests *(2024)*

Work Experience

Adobe | *Software Engineering Intern* *(May 2026 - Jul 2026)*
Noida, India

- Worked on **color-profile conversion** using **WGSL compute shaders**, enabling efficient GPU-based processing
- Developed a **WebGPU-based graphics implementation**, using modern GPU APIs for efficient web-based rendering and processing, with improved **performance and compatibility across modern web browsers**
- Optimized **GPU-based graphics processing** through **parallel execution** and key **performance improvements**
- Gained practical experience in **GPU programming, color management, WGSL, WebGPU, debugging, and graphics optimization** while contributing to **production-level software and performance-focused apps**

Key Projects

Facial Expression Analysis | *Course Project: Digital Image Processing* *(Aug 2025 - Nov 2025)*
Guided by Prof. Suyash P. Awate

- Developed a **VGG-style CNN from scratch** for 7-class facial-expression recognition using **35,887 FER-2013 48×48 grayscale images**, with normalized preprocessing, dropout regularization, and no prior transfer learning
- Achieved **67.97% test accuracy** and **67.55% validation accuracy** across seven classes during model evaluation
- Analyzed model performance across training, validation, and test sets, with a final **78.63% training accuracy**
- Published the trained model as a **public Kaggle Model Card**, recording **40+ downloads in the last 9 months**

Chess Engine | *Self Project* *(Mar 2026 - Apr 2026)*

- Developed a **C++ chess engine** using **SFML**, **data structures**, and **algorithms** for game-state management
- Implemented **Minimax-based move selection** with validation for **castling, en passant, and pawn promotion**
- Designed a position-evaluation system using **heuristic features and weighted scoring**, incorporating material, positional, and game-state factors, along with **piece-square tables and mobility analysis** to rank **moves**
- Achieved **2088 Bullet** and **2040 Blitz** ratings across **40+ games each** on Lichess with the engine bot, demonstrating consistent playing strength across multiple **time controls** and competitive game settings through automated gameplay

Adaptive Learning Tutor (ALT) | *Course Project: Adaptive Tutoring System* *(Jan 2026 - Apr 2026)*
Guided by Prof. Ramkumar Rajendran

- Developed and deployed an **adaptive tutoring system** that personalizes learning based on **learner knowledge**
- Implemented **Bayesian Knowledge Tracing (BKT)** to estimate concept mastery from learner responses, tracking evolving knowledge states to personalize learning, monitor progress, and guide subsequent question selection effectively
- Designed a policy-based question selection mechanism that adapts **difficulty, question type, hints, and explanation depth** according to the learner's evolving knowledge state, performance history, and learning progress
- Integrated learner performance and behavioral signals into the personalization pipeline to support adaptive learning

Multi-View 3D Object Reconstruction | *Course Project: Computer Vision* *(Jan 2026 - Apr 2026)*
Guided by Prof. Sharat Chandran

- Developed a deep-learning pipeline for **3D object reconstruction from multiple 2D viewpoints** using a lightweight multi-view feature extraction approach to learn **coarse geometric object representations** effectively
- Aggregated multi-view features and passed them to a **3D volumetric decoder** to predict coarse object geometry
- Trained the model using **binary cross-entropy loss** on a **custom multi-view object dataset**, and evaluated reconstructed 3D geometry using **voxel-based reconstruction accuracy** across held-out validation samples
- Conducted experiments with varying numbers of input views to study their effect on **3D reconstruction quality and model performance**, comparing **reconstruction accuracy and consistency** across different view configurations

Mithila-Lipi | *Self Project* *(Aug 2025 - Sep 2025)*

- Developed a browser-based **Devanagari → Tirhuta (Mithilakshar) script converter** using client-side Unicode-based conversion, enabling seamless **real-time script transliteration** directly within the browser environment
- Added an English → Maithili → Tirhuta workflow to support broader **multilingual language-to-script conversion**
- Reached **15K+ website impressions** with a **4+% average CTR**, exceeding typical classroom project engagement

EnvVault | *Self Project* *(Jun 2025 - Aug 2025)*

- Developed a lightweight **Linux Python virtual-environment manager** using Python's native `venv` framework
- Designed a **space-efficient storage architecture** with automatic archival and on-demand restoration of environments, alongside **environment-aware prompts and command/tab completion** for streamlined developer workflows
- Implemented environment lifecycle operations including **creation, activation, deactivation, listing, and removal** with Bash shell integration, providing **simple command-line management** for Python development environments
- Built installation and uninstallation scripts for a **reproducible and portable** command-line developer-tool setup

Additional Projects

Bash Autograder & VCS | *Course Project: Software Systems Lab* *(Jan 2024 - Apr 2024)*

Guided by Prof. Kameswari Chebrolu

- Built a command-line autograding and CSV-processing system using **Bash and Python**, with automated score aggregation, **result validation**, and streamlined data processing workflows for **efficient evaluation and reporting**
- Implemented a **Git-inspired version-control workflow** supporting `init`, `commit`, `checkout`, and `log` operations

Plagiarism Checker | *Course Project: Data Structures & Algorithms* *(Aug 2024 - Nov 2024)*

Guided by Prof. Ashutosh Gupta

- Developed a source-code similarity detector using **Levenshtein distance** and **Longest Common Subsequence (LCS)** to identify potentially plagiarized submissions through **structural and textual similarity techniques**
- Structured the application for reproducible execution using **Docker**, ensuring consistency across different systems

xv6 Kernel Enhancements | *Course Project: Operating System* *(Jan 2025 - Apr 2025)*

Guided by Prof. Mythili Vutukuru

- Extended the **xv6 operating system** with **demand paging, copy-on-write fork, and weighted round-robin scheduling**, enabling efficient **memory management, process scheduling, and overall resource utilization**
- Improved **kernel-level resource allocation and CPU scheduling** for more efficient process execution under load

Mentorship & Leadership

Dakshana Examination | *Examination Coordinator* *(Dec 2025)*

- Coordinated and invigilated the **Dakshana examination** at **JNV Deoghar** for **120+ students over two days**, as part of a nationwide assessment with **50,000+ participants across districts** for JEE and NEET preparation
- Managed examination-day supervision, attendance verification, and **secure conduct of the assessment** over two days, ensuring strict adherence to examination protocols and a **smooth and professional testing environment**

JoSAA Counselling Mentorship | *Student Mentor* *(Jun 2024 - Jul 2024)*

- Mentored students during the **JoSAA counselling**, assisting them with institute and branch selection decisions
- Guided students in evaluating **college, branch, and preference choices** based on their ranks, interests, and career goals, helping them make **informed decisions and effectively optimize their JoSAA preference order**

Courses Undertaken

- **Computer Science**
  Computer Programming and Utilization, Discrete Structures, †Software Systems Lab, †Data Structures and Algorithms, Logic for Computer Science, †Digital Logic Design and Computer Architecture, Design and Analysis of Algorithms, †Operating Systems, Automata Theory, †Abstractions and Paradigms for Programming, †Computer Networks, Game Theory and Algorithmic Mechanism Design, †Implementation of Programming Languages, †Database and Information Systems, Computer Graphics*
- **Data Science & Mathematics**
  Data Analysis and Interpretation, †Artificial Intelligence and Machine Learning, Fundamentals of Digital Image Processing, Computer Vision, Adaptive Tutoring Systems, Calculus, Linear Algebra and Differential Equations, Mathematics for Machine Learning: Optimization and Linear Algebra*

†*Course has corresponding lab* **Ongoing course*

Technical Skills

Languages	C++, C, Python, JavaScript, Bash, SQL, Rust, MIPS Assembly
ML / Data	NumPy, Pandas, Scikit-learn, TensorFlow, Keras, PyTorch, OpenCV
Systems & Web	Linux, Git, sed, awk, Docker, WebGPU, WGSL, SFML, React, Node.js, Django, HTML, CSS

Extracurricular Activities

- Participated in the **National Chess Competition** organized by **Navodaya Vidyalaya Samiti (NVS)**
- Participated in **NSO Chess at IIT Bombay**, competing in chess activities and inter-institute sporting events
- Participated in **IICPC**, gaining experience in algorithmic problem-solving through competitive programming contests
- Participated actively in **Seasons of Code (SoC)** organized by the **Web and Coding Club, IIT Bombay**